SoK: Hash-Based Polynomial Commitments and Low-Degree Tests
From FRI to Basefold, STIR, and WHIR

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Abstract. Hash-based low-degree tests and polynomial commitment schemes have become the cryptographic engine of a large fraction of deployed succinct-argument systems. Unlike pairing-based commitments such as KZG, they require no trusted setup, rely only on a collision-resistant hash, are plausibly post-quantum, and operate over *any* sufficiently large field, unlocking small fields whose arithmetic is fast on commodity hardware. Between 2017 and 2025 this design space evolved rapidly along two intertwined lineages: the Reed–Solomon *proximity-test* line (FRI, DEEP-FRI, STIR, WHIR) and the linear-code *tensor-commitment* line (Ligero, Brakedown), which Basefold and WHIR ultimately merge. Yet the primary sources report their guarantees under different, and frequently conflated, soundness regimes: unique decoding, the Johnson list-decoding bound, and (conjecturally) capacity. The folklore surrounding these schemes has drifted from what the papers actually prove.

We systematize this line of work. We organize the seven core schemes along a taxonomy of committed object, code class, and testing mechanism; we trace the evolutionary chain in which each scheme answers a concrete limitation of its predecessor; and we ground the theory in a survey of production zero-knowledge systems, showing how field choice and soundness regime jointly explain real engineering decisions, including a sumcheck-based multilinear prover reaching mainnet in 2026. We make two systematizing corrections. First, the DEEP folklore: the out-of-domain trick that survives in deployment (DEEP-ALI, on the constraint side) is distinct from the low-degree-test modification it is usually conflated with. That modification was superseded for FRI soundness by the *Proximity Gaps* analysis, which also proves Johnson-bound FRI soundness and is itself routinely misattributed to the original FRI paper. Second, and more consequential: the capacity-soundness conjectures on which nearly all deployed systems set their parameters had their strongest, up-to-capacity forms, including the mutual-correlated-agreement conjecture behind the newest schemes, disproved over large fields in late 2025. Soundness up to the Johnson bound is unaffected and the practical repricing is modest, but the discount the ecosystem had tacitly taken was, in its optimistic form, wrong. Our central thesis is that this proven-versus-conjectured soundness axis, not asymptotic query complexity, is the load-bearing and

least-consistently-reported dimension of the design space, and the field has now had to reprice it.

Keywords: low-degree testing · polynomial commitments · FRI · STIR · WHIR · Basefold · Reed–Solomon proximity · IOP · transparent SNARK · post-quantum · proximity gaps · soundness

1 Introduction

1.1 Motivation

A succinct non-interactive argument (SNARK) lets a prover convince a verifier that it knows a witness to an NP statement, using a proof far shorter than the witness and checkable far faster than re-execution. Almost every efficient SNARK factors into two components. The first is an *information-theoretic* protocol, a polynomial interactive oracle proof (PIOP) or algebraic IOP that reduces circuit satisfiability to a small number of polynomial identity checks. The second is a *cryptographic* component, a polynomial commitment scheme (PCS) that lets the prover commit to polynomials and later open evaluations, binding it to a fixed polynomial. The PCS is where the two dominant families of modern proof systems diverge.

One family builds the PCS from *pairings*: KZG commitments [17] give constant-size proofs and constant-time verification, but require a structured reference string (a trusted setup) and a pairing-friendly elliptic curve, hence a field of roughly 256 bits, and offer no post-quantum security. The other family builds the PCS from *hashing*: the prover Merkle-commits the evaluation of a polynomial over an error-correcting code, and a low-degree test or proximity test convinces the verifier that the committed word is close to a valid codeword. This second family, the subject of this systematization, needs no trusted setup, only a collision-resistant hash, is plausibly secure against quantum adversaries, and works over *any* sufficiently large finite field.

That last property is not incidental. It is the reason the hash-based family dominates high-throughput proving. Because the field is unconstrained, implementers choose fields whose arithmetic maps onto a CPU or GPU word: the 64-bit Goldilocks field ($p = 2^{64} - 2^{32} + 1$), the 31-bit BabyBear ($2^{31} - 2^{27} + 1$) and KoalaBear ($2^{31} - 2^{24} + 1$) fields, the Mersenne-31 field ($2^{31} - 1$), or towers of binary fields. Polygon reports that moving from a 256-bit KZG field to 64-bit Goldilocks with a FRI commitment yielded roughly a 40× proving speedup [18]. Field agility is the hash-based family's structural advantage, and, as we argue in Section 6, it explains the deployment landscape more cleanly than any other factor.

1.2 Why systematize now

The hash-based PCS design space has moved fast. FRI [2] was published in 2018, DEEP-FRI [5] in 2019, and the *Proximity Gaps* analysis that underpins

modern FRI soundness [3] in 2020; Basefold [9], STIR [7], and WHIR [8] all appeared between 2023 and 2025. Over the same window the parallel linear-code lineage (Ligero [1], Brakedown [12]) was absorbed into the folding lineage by Basefold. New soundness analyses, such as Haböck's list-decoding treatment of Basefold [10], and new *negative* results on the governing conjectures, including the late-2025 disproof of the up-to-capacity conjectures over large fields (§5.4), continue to appear.

The result is a literature that is at once mature enough to deploy, since these schemes secure substantial on-chain value, and unstable enough that its central guarantees are stated inconsistently across sources. Three specific problems motivate a systematization:

1. **Regime conflation.** Each scheme's soundness is proven at a particular *decoding radius*: the unique-decoding radius $(1-\rho)/2$, the Johnson (list-decoding) radius $1-\sqrt{\rho}$, or the conjectured capacity radius $1-\rho$. Papers, blog posts, and even other surveys routinely quote a scheme's *conjectured* parameters as though they were proven, or attribute a bound to the wrong source. The practical gap is large: at rate $1/4$ and 80-bit security, ethSTARK's own analysis needs 31 FRI queries under the capacity conjecture but 79 under the proven Johnson analysis, a $2.5\times$ difference in proof size [13].
2. **Two lineages, one destination.** The proximity-test line (FRI-descended) and the tensor-commitment line (Ligero-descended) are usually surveyed separately, as though they were unrelated. They are not unrelated. Basefold and WHIR are the points where the two lineages merge, and that convergence is the central structural fact in the recent literature.
3. **Theory–deployment gap.** Which production system uses which scheme, over which field, in which soundness regime, and *why*, is scattered across engineering blogs and specifications. No single source connects the asymptotic and concrete claims of the papers to the choices real systems make.

1.3 Contributions

This paper makes the following contributions, organized as three layered lenses on the same body of work.

- **A comparative taxonomy (Section 3, the spine).** We classify the seven core schemes along three axes: committed object (univariate vs. multilinear), code class (Reed–Solomon, general linear, foldable), and testing mechanism (folding IOPP vs. tensor consistency). For each scheme we give head-to-head tables on query complexity, proof size, prover cost, verifier cost, and soundness radius, with every cell attributed to a primary source and tagged by regime.
- **An evolutionary narrative (Sections 4–5, the connective tissue).** We trace the causal chain in which each scheme is an engineered response to a specific, identifiable limitation of its predecessor: DEEP-FRI answers FRI's soundness ceiling; Brakedown answers Ligero's field-dependence; Basefold answers the field-agnostic-yet-succinct gap and the univariate–multilinear

mismatch; STIR answers FRI's constant-rate query barrier; WHIR answers STIR's verification cost and unifies the univariate and multilinear worlds.
– **A deployment-grounded analysis (Section 6, the payoff).** We map more than a dozen production and near-production systems to the schemes they use, and argue that two coupled variables, field choice and soundness regime, explain the observed engineering decisions, including the common "fast FRI prover, KZG/Groth16 final wrapper" hybrid pattern.

Cutting across all three, we surface several places where the community's understanding has drifted from the primary sources. The most substantive is a distinction the folklore routinely blurs: the DEEP *low-degree-test* modification, largely superseded for FRI soundness by [3], versus **DEEP-ALI**, the out-of-domain quotient on the constraint side that actually ships in deployed STARKs (§4.2). We also flag that Basefold's field-agnostic construction is proven only at unique decoding, while its list-decoding guarantee [10] applies to the Reed–Solomon instantiation (§5.2). A smaller but persistent attribution error we correct is that Johnson-bound FRI soundness belongs to [3], not to the FRI paper. Finally, we document that the capacity-soundness conjectures underpinning deployed parameters are, as of 2026, under active revision rather than settled, a point whose scope we treat carefully in §5.4 rather than overstate.

1.4 Scope and non-goals

We restrict attention to *hash-based* (Merkle-committed, transparent) low-degree and proximity tests and the polynomial commitments built directly on them. We treat pairing-based (KZG) and discrete-log (IPA/Bulletproofs, Dory) commitments only as points of contrast in Section 6. We do not cover the *front-end* PIOPs (PLONK, Spartan, HyperPlonk) except insofar as a scheme's committed object determines which PIOPs it can serve. In this paper "folding" always means the FRI-style degree-halving of a codeword; we do **not** treat the Nova/ProtoStar line of *accumulation/folding schemes*, which fold NP instances rather than codewords and lie outside our scope. We do not reprove any theorem; our claims about each scheme are claims about *what its source reports*, and we are explicit throughout about the difference between a proven and a conjectured statement.

1.5 Relation to prior systematizations

Three existing works cover parts of this territory, and delineating our delta from them is essential to justifying a new systematization.

Haböck, "A Summary on the FRI Low Degree Test" [14] is the closest prior work and an excellent reference. It systematizes FRI soundness (proximity gaps, correlated agreement, the proven-versus-conjectured query counts) and is a source we build on directly in Section 4. But its scope is deliberately narrow: it treats FRI (and its soundness) alone. It does not cover the tensor lineage (Ligero, Brakedown), the foldable-code generalization (Basefold), the rate-reduction and constrained-code successors (STIR, WHIR), the univariate–multilinear merger

that is our central structural claim, or any deployment mapping. Our paper's reason to exist is what [14] scopes out: the *cross-lineage* view and its convergence.

Thaler, "Proofs, Arguments, and Zero-Knowledge" (2022) [21] is the standard textbook and covers the tensor line (Ligero, Brakedown) and FRI pedagogically and well. As a textbook predating Basefold, STIR, and WHIR, it neither includes the 2023–2025 mergers nor frames the proven-versus-conjectured soundness regime as an organizing axis; it teaches the schemes, it does not systematize the design space around a thesis.

Chiesa–Yogev, "Building Cryptographic Proofs from Hash Functions" (2024) [22] is the authoritative monograph on the IOP-to-argument compilation (BCS, round-by-round soundness, Fiat–Shamir in the ROM) that our Section 2.3 and the machinery of Section 5.5 depend on. It is a foundations reference for *how* one obtains an argument from any IOP; it is not a comparative survey of the specific low-degree tests, their concrete costs, or their deployment.

Our contribution relative to all three is the combination that none provides: (i) both lineages and their Basefold/WHIR merger under one taxonomy; (ii) the proven-versus-conjectured soundness regime as the explicit organizing axis, carried through concrete tables; and (iii) a sourced deployment map connecting the theory to production engineering.

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2 Preliminaries

2.1 Reed–Solomon codes and proximity

Fix a finite field $\mathbb{F}$, an evaluation domain $S \subseteq \mathbb{F}$ with $|S| = n$, and a rate $\rho \in (0, 1)$. The Reed–Solomon code $\mathrm{RS}[\mathbb{F}, S, \rho]$ is the set of evaluation vectors $(p(x))_{x \in S}$ of polynomials p of degree $d < \rho n$. Its rate is $\rho = d/n$ (up to the additive one), and because Reed–Solomon codes are maximum-distance-separable (MDS), their relative minimum distance is $1 - \rho + 1/n \approx 1 - \rho$. The *blowup* factor is $1/\rho$.

The relative Hamming distance $\Delta(w, C)$ between a word $w \in \mathbb{F}^n$ and a code C is the minimum fraction of coordinates in which w differs from any codeword. A *low-degree test* (equivalently, an RS *proximity test*) is a protocol that, given oracle access to w, distinguishes the case $w \in C$ (completeness) from the case $\Delta(w, C) \geq \delta$ (soundness), while querying w in few coordinates. The parameter δ is the *proximity parameter*, and the *decoding radius* at which a test is analyzed is the central quantity of this paper.

2.2 Three decoding radii

Reed–Solomon proximity behaves qualitatively differently in three regimes, defined by how far δ is pushed toward the code's distance $1 - \rho$:

- **Unique decoding**, $\delta < (1 - \rho)/2$. Below half the minimum distance, every word has at most one nearby codeword; extraction and binding arguments are simplest here.

- **Johnson / list-decoding bound**, $(1-\rho)/2 \leq \delta < 1-\sqrt{\rho}$. Up to the Johnson radius $1-\sqrt{\rho}$, Reed–Solomon codes are list-decodable with *polynomially bounded* list size (a classical result; the Guruswami–Sudan regime). Most modern proximity-test soundness proofs live here.
- **Capacity**, $\delta < 1-\rho$. Every scheme in this survey has a *conjectured* soundness statement in this regime, and nearly every deployed system sets parameters as though the conjecture holds. The status of this regime changed sharply in November 2025: the strongest *up-to-capacity* conjectures (small list size, correlated agreement, and mutual correlated agreement all the way to $1-\rho$) were **disproved over large fields** by Crites and Stewart [24], with a matching exception-count barrier at the Johnson boundary from [25]. We foreground this throughout (see §5.4); for now the reader should treat every "capacity" entry below as *conjectural and, in its strongest form, now refuted*, with soundness up to the Johnson bound unaffected.

The distinction is not academic. Under a capacity-regime analysis, each proximity query "buys" roughly $\log_2(1/\rho)$ bits of soundness; under the proven Johnson analysis, roughly half that. Query count, and therefore proof size and verifier hashing, scales inversely, so choosing to trust the capacity conjecture roughly halves the proof.

2.3 From IOP to non-interactive argument

All schemes here are first described as *interactive oracle proofs* (IOPs) or IOPs *of proximity* (IOPPs): the prover sends oracles (long strings), the verifier queries a few positions and sends random challenges. The BCS transformation [4] compiles any public-coin IOP into a non-interactive argument in the random-oracle model by Merkle-committing each oracle (the commitment is the root; an opening is an authentication path) and deriving the verifier's challenges via Fiat–Shamir. Consequently, three IOP quantities map directly to argument costs: the number of oracle *queries* becomes the number of Merkle authentication paths (the dominant proof-size and verifier-hashing cost); the *round* count becomes the number of committed oracles; and the prover's arithmetic plus hashing becomes prover time. Throughout, "query complexity" is the quantity to minimize, because it multiplies the per-path hash cost.

2.4 The proximity-gaps engine

A recurring technical device deserves early mention because it silently underlies FRI, STIR, WHIR, and the list-decoding analysis of Basefold. A code C exhibits a (δ, ε)-*proximity gap* with respect to a family of words (typically an affine or linear space) if every such space is either (i) entirely within relative distance δ of C, or (ii) has at most an ε fraction of its members within distance δ: there is no space that is "half close." *Correlated agreement*, the companion notion, states that when a random linear combination of words is close to the code, the individual words agree with a codeword on a common large coordinate set. These

results, proven for Reed–Solomon codes up to the Johnson bound by Ben-Sasson, Carmon, Ishai, Kopparty, and Saraf [3], are what let a proximity test *batch* many words into one and *fold* a codeword while preserving distance. When we say a scheme's soundness "reaches the Johnson bound," we almost always mean it inherits [3]. We now make these statements formal.

2.5 Formal statements

Because the paper's thesis concerns soundness, we state the central technical objects formally rather than only in prose. Let $\mathrm{RS} = \mathrm{RS}[\mathbb{F}, S, \rho]$ with $|S| = n$ and $q = |\mathbb{F}|$.

Definition 1 (decoding radii). For $w \in \mathbb{F}^S$, the *unique-decoding radius* is $(1-\rho)/2$; the *Johnson (list-decoding) radius* is $J_\rho := 1-\sqrt{\rho}$; the *capacity radius* is $1-\rho$. For $\delta < J_\rho$, Reed–Solomon codes are (δ, L)-list-decodable with $L = \mathrm{poly}(n)$ (Guruswami–Sudan); the hope that a small list size persists for $\delta \in [J_\rho, 1-\rho)$ was **disproved over large fields** in late 2025 [24] (§5.4), removing the support for the up-to-capacity soundness conjectures.

Theorem 1 (proximity gaps for RS [3, Thm. 1.2], informal). Let $U = \{u + zv : z \in \mathbb{F}\}$ be an affine line in $\mathbb{F}^S$ (the statement holds for affine spaces generally). For every proximity parameter $\delta \in (0, 1-\sqrt{\rho})$, either $\Delta(u', \mathrm{RS}) \le \delta$ for *every* $u' \in U$, or $\Pr_z[\Delta(u+zv, \mathrm{RS}) \le \delta] \le \varepsilon$, where

$$\varepsilon = \varepsilon(\delta, n, q, m) = \begin{cases} \dfrac{n}{q}, & \delta \in \left(0, \frac{1-\rho}{2}\right] \quad \text{(unique-decoding regime)}, \\ \dfrac{(m+\frac{1}{2})^7}{3\,\rho^{3/2}} \cdot \dfrac{n^2}{q}, & \delta < 1-\sqrt{\rho}\left(1+\frac{1}{2m}\right),\ m \in \mathbb{Z}_{\ge 3} \quad \text{(Johnson regime)}. \end{cases}$$

Here $m \ge 3$ is a free integer parameter: as $m \to \infty$ the reachable radius $1-\sqrt{\rho}\,(1+\frac{1}{2m})$ approaches the Johnson bound $1-\sqrt{\rho}$ from below while the error grows as $(m+\frac{1}{2})^7$. This is the same m that appears in the per-query factor $\sqrt{\rho}\,(1+\frac{1}{2m})$ of the Corollary below.

Theorem 2 (correlated agreement [3, Thm. 1.4], informal). Under the hypothesis of Theorem 1 case (ii)'s negation, i.e. when a $> \varepsilon$ fraction of U is δ-close to RS, there is a *single* set $S' \subseteq S$ with $|S'| \ge (1-\delta)n$ and codewords $u^*, v^* \in \mathrm{RS}$ such that $u|_{S'} = u^*|_{S'}$ and $v|_{S'} = v^*|_{S'}$. That is, the closeness is *correlated*: the words agree with the code on a *common* large domain, not individually.

Corollary (batched FRI soundness [3, Thm. 8.3], informal). For FRI with r folding rounds and ℓ query repetitions over a rate-ρ code, a prover whose input is δ-far from RS is accepted with probability at most

$$\varepsilon_{\mathrm{FRI}} \;\le\; \varepsilon_{\mathrm{C}} \;+\; \left(\sqrt{\rho}\left(1+\tfrac{1}{2m}\right)\right)^{\ell},$$

where $\varepsilon_{\mathrm{C}} = O\big(\log n \cdot m^7 n^2/(\rho^{3/2} q)\big)$ collects the round-consistency error summed over the $O(\log n)$ folding rounds. The per-query factor $\to \sqrt{\rho}$ as $m \to \infty$, i.e. each query contributes $\approx \frac{1}{2}\log_2(1/\rho)$ bits of soundness in the *proven* (Johnson) regime,

versus $\log_2(1/\rho)$ bits under the *conjectured* capacity error ρ^ℓ. This factor-of-two in per-query soundness is the formal root of the 2–2.5× concrete query gap tabulated in Section 5.4.

Definition 2 (constrained Reed–Solomon code [8, Def. 1]). For a weight polynomial $\hat{w}$ and target $\sigma \in \mathbb{F}$,

$$\mathrm{CRS}[\mathbb{F}, S, m, \hat{w}, \sigma] := \Big\{ f \in \mathrm{RS}[\mathbb{F}, S, m] : \textstyle\sum_{b \in \{0,1\}^m} \hat{w}\big(\hat{f}(b), b\big) = \sigma \Big\},$$

where here $\mathrm{RS}[\mathbb{F}, S, m]$ denotes evaluations on S of polynomials of degree $< 2^m$ (the third slot is WHIR's number-of-variables parameter, not a rate), and $\hat{f}$ is the multilinear extension of f's message. A single such constraint can encode a multilinear evaluation, a univariate evaluation, or an inner-product claim; this expressiveness is what lets WHIR subsume both the FRI/STIR (univariate) and Basefold (multilinear) query types.

Definition 3 (mutual correlated agreement [8], informal). A strengthening of Theorem 2: for words $f_1, \ldots, f_\ell$ and any set $S' \subseteq S$ with $|S'| \geq (1-\delta)n$, if the random combination $\sum_i \alpha_i f_i$ agrees with RS on S' with high probability over the α_i, then *every* f_i individually agrees with RS on the *same* S'. The original WHIR analysis proves this in the unique-decoding regime and *conjectures* it up to capacity, which drives WHIR's concrete parameters. Subsequent work of Haböck [11] proves mutual correlated agreement for Reed–Solomon codes beyond unique decoding, up to the Johnson list-decoding bound, in line with the proximity-gap results of [25].

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3 A Taxonomy of Hash-Based Commitments

3.1 Three classifying axes

The seven schemes are most usefully separated along three axes.

Axis 1: Committed object. Does the scheme commit to a *univariate* polynomial (FRI, DEEP-FRI, STIR; Ligero in its basic form) or to a *multilinear* polynomial over the Boolean hypercube (Basefold, WHIR, Brakedown's polynomial-commitment use)? This axis governs *compatibility*: univariate commitments serve univariate PIOPs and AIR-based STARKs, while multilinear commitments serve sumcheck-based PIOPs such as HyperPlonk and Spartan. A scheme that commits to the "wrong" object must pay for an adapter.

Axis 2: Code class. Reed–Solomon (FRI, DEEP-FRI, STIR, WHIR, Ligero), a general linear-time-encodable code (Brakedown), or a *foldable* linear code (Basefold's random foldable codes; WHIR's constrained Reed–Solomon). The code class determines the *field requirement* (Reed–Solomon wants FFT-friendly smooth-order fields; expander and foldable codes are field-agnostic) and the *prover encoding cost* ($O(n \log n)$ for FFT, $O(n)$ for linear-time codes).

Axis 3: Testing mechanism. *Folding* (FRI-style: recursively halve the degree by combining evaluation points, one Merkle oracle per round) versus

tensor consistency (Ligero/Brakedown-style: arrange coefficients in a matrix, encode rows, test a random linear combination of rows against opened columns). Folding gives polylogarithmic proofs at the cost of $\log n$ rounds and oracles; the one-shot tensor test gives a $\sqrt{n}$ proof in a single round.

3.2 The two lineages and their merger

These axes induce two lineages, shown in Figure 1 (conceptual).

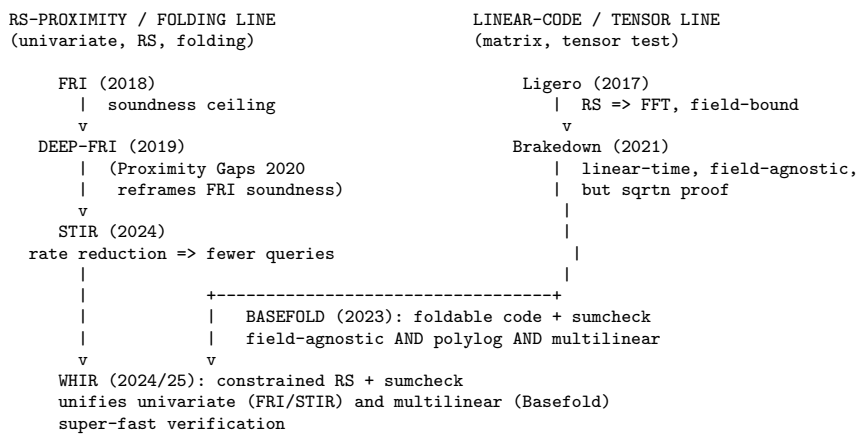


Figure 1. The two lineages (folding and tensor) and their merger points, Basefold and WHIR. Conceptual; not to scale.

The left column is the folding line: FRI and its soundness-improving descendant DEEP-FRI, then STIR's query-reducing rate schedule. The right column is the tensor line: Ligero and its field-agnostic, linear-time successor Brakedown. **Basefold (2023) is the first merger point**: it takes the folding mechanism of the left column but runs it over a field-agnostic code in the spirit of the right column, and adds sumcheck to commit to multilinear polynomials. **WHIR (2024/25) is the second and more complete merger**: its constrained Reed–Solomon codes express both univariate and multilinear queries, so it is simultaneously a successor to STIR (rate reduction, few queries) and to Basefold (multilinear, sumcheck), with much faster verification than either.

3.3 Master comparison table

Table 1 is the spine of the systematization. Every cell is attributed; soundness columns separate *proven* from *conjectured*. N denotes witness/circuit size, d degree, n codeword length, ρ rate, λ security bits, and ℓ/t query repetitions.

Table 1: Core schemes across the comparison axes.

Scheme	Object	Code	Query complexity	Proof size	Prover	Verifier	Proven radius	Conjectured
FRI [2]	univar.	RS	$2\log N + \ell$ reps, $\ell \approx 2\lambda/\log\frac{1}{\rho}$ (via [3])	$O(\log^2 N)$	$O(N \log N)$ †	$O(\log N)$ arith. + hashing ‡	$\delta_0 \approx \frac{1-3\rho}{4}$ (orig.); $1-\sqrt{\rho}$ via [3]	capacity $1-\rho$ (refuted §5.4)
DEEP-FRI [5]	univar.	RS	$\ell \approx 3\lambda/\log\frac{1}{\rho}$ (tight at $q=n$)	$O(\log^2 N)$	$O(N \log N)$ †	$O(\log N)$ + hashing ‡	$1-\sqrt{\rho}$ (Johnson)	capacity $1-\rho$ (refuted §5.4)
Ligero [1]	(circuit-SAT; not native PCS)	RS	$t = \Theta(\lambda)$ cols	$O(\sqrt{N})$	$O(N \log N)$ (FFT)	$O(N)$	$e < d/4$ (orig.); $d/3$ later	n/a
Brakedown [12]	multilin.	expander	$t = \Theta(\lambda)$ cols (more than Ligero)	$O(\sqrt{N})$, tens of MB	$O(N)$, no FFT, field-agnostic	$O(\sqrt{N})$	code-distance (unique-dec.)	n/a
Basefold [9]	multilin.	foldable (RFC)	ℓ reps	$O(\log^2 N)$	$O(N \log N)$ enc, $O(N)$ eval	$O(\log^2 N)$	unique-dec. (orig.); $1-\sqrt{\rho}$ for RS via [10]	capacity (refuted §5.4)
STIR [7]	univar.	RS (rate reduced)	$O(\log d + \lambda \log\log d)$	1.25–2.46× < FRI	$O(n \log n)$	∼FRI	$1-\sqrt{\rho}$ (Johnson)	capacity $1-\rho$ (refuted §5.4)
WHIR [8]	both	constrained RS	$O(\log n + \lambda \log\log n)$	36–187 KiB	$O(n \log n)$	210–770 μs (100-bit)	Johnson (proximity); unique-dec. mutual corr. agr. (§5.3)	capacity $1-\rho$ (refuted §5.4)

† **Prover columns are normalized to *total* cost including the low-degree extension.** FRI's celebrated "$< 6n$" figure [2, Thm. 2] is the *folding-phase arithmetic only*, given the input codeword; producing the initial Reed–Solomon oracle is an $O(N \log N)$ FFT, just as for STIR/WHIR/Basefold. FRI does not have an asymptotically cheaper prover than STIR/WHIR. ‡ **Verifier arithmetic** is logarithmic ($\sim 21 \log N$ ops for FRI [2]), but the *dominant* verifier cost after BCS compilation is Merkle-path **hashing**, which scales with query count × path length (§2.3), the quantity the query-complexity column ultimately governs.

Two reading notes. First, the *query complexity* column is where STIR and WHIR make their headline advance: FRI's $\Theta(\lambda \log d)$ becomes STIR/WHIR's $O(\log d + \lambda \log\log d)$. Second, the *proof size* column mixes asymptotic and concrete entries deliberately, because the schemes are usually compared concretely; the linear-code schemes' $O(\sqrt{N})$ is asymptotically worse than the folding schemes' $O(\log^2 N)$ but comes from a single-round, simple protocol, which is why Ligero and Brakedown remain attractive in low-memory settings (Section 6).

3.4 What the taxonomy reveals

Three observations fall out of Table 1 that the per-scheme literature tends to obscure.

The field-agnostic frontier moved. Ligero and FRI both need FFT-friendly fields (Reed–Solomon encoding). Brakedown broke field-dependence but paid with $\sqrt{N}$ proofs. Basefold is the first scheme to be simultaneously field-agnostic *and* polylogarithmic: it dominates Brakedown on proof size by more than $10\times$ at scale [9] while retaining field agility. This is the quiet but important structural advance of 2023.

Query complexity and proof size decoupled from prover time. STIR and WHIR reduce query complexity (and thus proof size and verifier hashing) *without* increasing prover time relative to FRI: the prover remains $O(n \log n)$, FFT-dominated. The improvement is purely in the information-theoretic efficiency of the test, extracted from the rate schedule, not a prover/verifier trade.

The soundness columns are where the schemes are least comparable. Every "conjectured" cell in Table 1 depends on an unproven capacity statement, and the proven cells sit at heterogeneous radii (unique decoding for Ligero, Brakedown, and original Basefold; Johnson for DEEP-FRI, STIR, and Basefold-over-RS). Comparing two schemes' proof sizes is only meaningful *within a fixed regime*, a point we develop in Section 5.

3.5 The axes are separable: placing the deployed variants

A fair objection to any taxonomy is that its axes may restate the historical narrative: here, that "Reed–Solomon $\Longrightarrow$ folding $\Longrightarrow$ univariate" (left lineage) and "linear code $\Longrightarrow$ tensor $\Longrightarrow$ matrix" (right lineage) co-vary so tightly that the three axes collapse to one. Two *deployed* variants that do not appear among the seven core schemes show the axes are independent, by occupying cells the lineage view does not predict.

Circle-FRI (the low-degree test inside Circle STARKs [15]) holds the *testing mechanism* axis fixed at FRI-style folding and the *object* axis at univariate, but changes only the *code*: from a standard Reed–Solomon code over a multiplicative coset to an evaluation code over the *circle group* $\{x^2 + y^2 = 1\}$ of a non-two-adic field (Mersenne-31). It is thus "FRI with a different code class", a point on the folding axis reachable only by moving the code axis alone, which the lineage story (code and mechanism moving together) does not generate.

FRI-Binius [16] holds the *mechanism* axis at Basefold-style folding-plus-sumcheck and the *object* axis at multilinear, but moves the *code/field* axis to towers of binary fields, where the encoding is an additive-NTT rather than a multiplicative FFT. It is the multilinear-foldable cell instantiated over a field class (characteristic two, no smooth multiplicative subgroup) that neither Basefold's random foldable codes nor WHIR's constrained Reed–Solomon codes assume.

These two variants, both in production or near-production (Section 6), sit at taxonomy coordinates that the two-lineage narrative alone would not enumerate, which is the operational sense in which the axes do independent work. They

also justify treating the taxonomy as a *method* for classifying future schemes (an incoming scheme is located by its three coordinates), not a snapshot of the current seven. We nonetheless concede that within the seven *core* schemes the axes are strongly correlated, and that the deployment lens (Section 6) is the layer contributing the most new organization.

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4 The Two Lineages Before the Merger: FRI to STIR, Ligero to Brakedown

This section and the next trace the evolutionary narrative: each scheme as an answer to a specific limitation of its predecessor. We treat the folding line here and the tensor line in Section 4.5, then the mergers.

4.1 FRI: strictly-linear folding, and a soundness ceiling

FRI (Fast Reed–Solomon IOP of Proximity) [2] tests proximity to $\mathrm{RS}[\mathbb{F}, S, \rho]$ by *folding*. Given a codeword-candidate of degree d over a domain of size n, the prover splits its polynomial into even and odd parts, $p(X) = p_e(X^2) + Xp_o(X^2)$, and the verifier's random challenge α combines them into $p_e + \alpha p_o$, a polynomial of degree $d/2$ over a domain of size $n/2$. Repeating for $\log d$ rounds reduces to a constant-size instance; each round the prover commits a Merkle root, and in the query phase the verifier checks consistency between consecutive rounds using two queries per invocation. The headline costs are exceptional: the *folding-phase* prover arithmetic is below $6n$ operations (strictly linear given the input codeword), the verifier is *strictly logarithmic* ($\sim 21 \log n$ operations), and the proof length is below $n/3$ before Merkle compilation [2, Thm. 2]. (The *total* prover, including the initial low-degree extension, is $O(n \log n)$ like every Reed–Solomon scheme here; the celebrated linearity is of the folding step, not the whole prover; cf. Table 1, note †.)

The limitation is soundness. FRI's original Theorem 2 proves a rejection probability governed by $\delta_0 \geq \frac{1}{4}(1 - 3\rho) - 1/\sqrt{n}$, a constant that sits roughly at *half* the unique-decoding radius, well short of the code's distance. The paper's abstract frames the guarantee as "negligible loss below the unique-decoding radius," and the full version pushes the analysis to the so-called double-Johnson radius $1 - (1 - \delta_V)^{1/4}$, but neither reaches the Johnson list-decoding radius. Because each query buys soundness proportional to the achievable distance, a low ceiling means many queries: the original analysis required a repetition count ℓ that did not even tend to zero error cleanly, later improved to $\ell \approx 4\lambda / \log(1/\rho)$ [6]. FRI also states a *capacity conjecture* (its Conjecture 3): that soundness in fact holds up to $1 - \rho$, which, if true, would minimize queries. This conjecture, not the theorem, is what deployed STARKs use.

> **Systematization flag 1 (miscitation).** The oft-repeated statement "FRI is sound up to the Johnson bound $1 - \sqrt{\rho}$" is *true* but is **not** proven

in [2]. It follows from the *Proximity Gaps* theorem [3] (Section 4.3), published two years later. Attributing Johnson-bound FRI soundness to the FRI paper (common in blog posts and even some peer-reviewed work) misplaces the result by two years and one paper.

4.2 DEEP-FRI: sampling outside the box

DEEP-FRI [5] targets FRI's soundness ceiling directly. The bottleneck FRI hit was a worst-case-to-average-case reduction: folding and random affine combinations could not push the *average* distance of a linear space past the double-Johnson bound $1-(1-\delta_V)^{1/4}$. DEEP-FRI's two contributions raise this.

First, an *analytic* improvement: the authors raise the general linear-code reduction to the "one-and-a-half Johnson" bound $1-(1-\delta_V)^{1/3}$ and prove it tight, which for Reed–Solomon translates to a per-query soundness error of $\max\{1-\delta, \rho^{1/3}\}$ and a 25% reduction in verifier complexity ($4\lambda \to 3\lambda$ over $\log(1/\rho)$). Second, and more influentially, the **DEEP method** ("Domain Extending for Eliminating Pretenders"): the verifier samples an evaluation point z *outside* the evaluation domain and asks the prover to reveal the interpolant's value there. Because $(X-z)$ divides $p(X)-p(z)$, an out-of-domain answer forces the prover to commit to at most one codeword from the list of nearby "pretenders," effectively pruning the list. With list size bounded (which holds below the Johnson bound), DEEP-FRI's main theorem yields proven soundness up to the **Johnson/list-decoding radius** $1-\sqrt{\rho}$, strictly better than FRI's proven ceiling, while retaining linear proving and logarithmic verification. DEEP-FRI too states a capacity conjecture (its Conjecture 2.3, that Reed–Solomon list size stays small up to $1-\rho$).

> **Systematization flag 2 (what actually survives).** The DEEP idea applied to the *low-degree test itself* was largely *superseded* for FRI soundness by [3], which showed *plain, unmodified* FRI already reaches $1-\sqrt{\rho}$ without the extra out-of-domain query in the folding phase. What survives in deployed systems is **DEEP-ALI**: applying out-of-domain sampling on the *constraint/quotient* side of the STARK (the "DEEP quotient"), which improves the algebraic-linking soundness from a small constant to nearly 1. Practitioners who speak of "DEEP" almost always mean this quotient, not a modified FRI fold.

4.3 Proximity Gaps: the analysis that reframed everything

The *Proximity Gaps for Reed–Solomon Codes* work [3] is not a new protocol but a new *analysis*, and it is the true foundation of modern FRI soundness. Its main theorem establishes that affine spaces exhibit a (δ, ε)-proximity gap with respect to Reed–Solomon codes for all δ up to the Johnson bound $1-\sqrt{\rho}$, with an error ε that is *linear* in n/q in the unique-decoding regime and *quadratic* (n^2/q) in the Johnson regime. Applied to FRI's batching and folding (its Theorem 8.3), this gives a per-query soundness factor approaching $\sqrt{\rho}$ and reduces the

required repetition count to $\ell \approx 2\lambda/\log(1/\rho)$ for $q \gg n^2$, a 33% shorter proof than DEEP-FRI's bound and a full halving relative to the earliest analysis. A worked example in the paper reaches 128 bits of *provable* soundness at rate 1/16 with $\ell = 65$ query invocations. [3] also states the capacity conjecture in its simplest form (its Conjecture 8.4).

The historical arc of FRI's query count is worth stating explicitly, because it shows that most of FRI's "improvement" came from *sharper analysis of the same protocol*, not new protocols: ℓ went from "large" [2] to $4\lambda/\log(1/\rho)$ [6] to $3\lambda/\log(1/\rho)$ [5, tight at $q = n$] to $2\lambda/\log(1/\rho)$ [3, for $q \gg n^2$].

4.4 STIR: reduce the rate, spend fewer queries

STIR ("Shift To Improve Rate") [7] is the first *protocol* change to FRI's folding since DEEP-FRI, and it attacks a different limitation: the **constant-rate barrier**. In FRI, each round folds the degree by a factor k *and* shrinks the domain by k, so the rate $\rho = d/n$ is held constant across all rounds. A constant rate means every round's proximity test is to a code of the same relative distance, so every query buys the same, capped amount of soundness, forcing $\Theta(\lambda \log d)$ total queries.

STIR breaks this by shrinking the domain *more slowly* than the degree (roughly by a factor of 2 rather than k) so that the rate *decreases* each round (new rate $\approx (2/k)\rho$). A lower-rate Reed–Solomon code has larger relative distance, so each successive round tests proximity to a "better" code and each query extracts strictly more soundness. To move between rounds while the rate falls, STIR uses out-of-domain (DEEP-style) sampling and degree-correction quotients. The payoff is a query complexity of $O(\log d + \lambda \log\log d)$ (replacing FRI's $\log d$ factor with $\log\log d$), which the authors state is the best known for any concretely efficient Reed–Solomon IOPP, *including* FRI variants that assume the capacity conjecture. Concretely, STIR yields arguments 1.25× to 2.46× smaller than a highly optimized FRI (e.g., 114 KiB vs. 211 KiB at 128-bit security, degree 2^{26}, rate 1/4), with prover and verifier times similar to FRI. STIR's proven soundness sits at the Johnson bound; like its predecessors it also offers conjectured capacity parameters.

4.5 The tensor line: Ligero and Brakedown

Before reaching the mergers we must introduce the second lineage, because Basefold and WHIR draw from it. Its conceptual root is the **tensor-query IOP** of Bootle–Chiesa–Groth [23]: a polynomial evaluation $q(z) = b^\top U a$ factors through a coefficient matrix U with query vectors a, b that are themselves tensor products, so an evaluation reduces to a random-linear-combination test on the rows plus a consistency check on the columns. Every scheme in this subsection is an instantiation of that template with a particular code.

Ligero [1] arranges the witness into an $m \times m$ matrix, encodes each *row* with a Reed–Solomon code to width n, and Merkle-commits the *columns* of the resulting $m \times n$ matrix. Two tests establish correctness: a *proximity/interleaved test* (a random linear combination of the rows should be a codeword, checked

by opening t random columns) and *consistency tests* for the circuit's linear and quadratic constraints on the same columns. The proof is $O(\sqrt{N})$ (a single-round, conceptually simple protocol needing only a hash and no trusted setup), and its soundness combines a field-collision term $\sim d/|\mathbb{F}|$ with a column-miss term $(1 - e/n)^t$, proven in the original CCS 2017 paper for $e < d/4$ (Lemma 4.2, Thm. 4.4). The sharper threshold $e < d/3$ was only *conjectured* by AHIV and later proven (via a Roth–Zémor argument, folded into the extended ePrint version), a small but representative instance of the proven-versus-later-tightened pattern that recurs throughout this survey. Ligero's limitation is inherited from Reed–Solomon: encoding costs $O(N \log N)$ FFTs and demands an FFT-friendly field.

Brakedown [12] keeps Ligero's tensor skeleton but swaps the code. Observing that Bootle–Chiesa–Groth's tensor-query IOP composes with Spartan's linear-time polynomial IOP, Brakedown instantiates the row code with a **linear-time-encodable expander code** (Spielman-style, from lossless expander graphs) that needs only a small *constant* relative distance and no field structure. The result is the first built SNARK with a **strictly linear-time, field-agnostic prover** ($O(N)$ field operations, no FFT), plausibly post-quantum and transparent. The price is the code's weaker distance: a small constant relative distance forces *more* column queries than Ligero's Reed–Solomon code, yielding the largest proofs among coding-based schemes (tens of MB at degree 2^{25}). The Brakedown paper also ships **Shockwave**, the same framework with Reed–Solomon encoding, trading field-agnosticism back for smaller proofs, a direct illustration of the field/proof-size tension.

Two intermediate schemes on this line attacked the $\sqrt{N}$ proof size *within* the tensor framework, before Basefold superseded the approach. **Ligero++** [20] keeps the interleaved-Reed–Solomon commit but replaces the $O(\sqrt{N})$ column-opening with a discrete-log/IPA-style inner-product argument, cutting the proof to polylogarithmic, at the cost of reintroducing public-key cryptography and forfeiting post-quantum security. **Orion** [19] (Xie–Zhang–Song, CRYPTO 2022) instead stays hash-based, adding an expander-testing step and a *code-switching* (proof-composition) layer to compress Brakedown-style proofs to $O(\log^2 N)$ (a few MB at degree 2^{25}) while retaining the linear-time expander-code prover. Orion is, in effect, the tensor line's own answer to "linear prover *and* polylog proof", reached contemporaneously with, but by a different route than, Basefold's foldable-code approach.

> **Evolutionary link.** Brakedown answers Ligero's precise limitation, Reed–Solomon's FFT and field-dependence, by generalizing the code to any linear-time-encodable family, buying field agility and a linear prover at the cost of proof size. Ligero++ and Orion then attack the $\sqrt{N}$ proof from inside the tensor framework (via IPA and via code-switching respectively). But it is **Basefold** that resolves the tension most cleanly, by importing the folding mechanism from the *other* lineage, which is why we treat it as a merger (§5.1) rather than a tensor-line descendant.

5 The Mergers: Basefold and WHIR, and the Soundness Axis

5.1 Basefold: folding meets field-agnosticism meets sumcheck

Basefold [9] is the first scheme to be simultaneously field-agnostic (like Brakedown) *and* polylogarithmic (like FRI) *and* multilinear. It has three moving parts.

First, it *generalizes FRI's folding* from Reed–Solomon to any **foldable linear code**: a code whose generator matrix has a recursive block structure ($G_i = [[G_{i-1}, G_{i-1}], [G_{i-1}T_{i-1}, G_{i-1}T'_{i-1}]]$ with the two diagonal families everywhere distinct) so that a codeword splits into left/right halves living under the *same* smaller code, which is what enables the FRI fold. Second, it introduces **random foldable codes (RFCs)**, a punctured Reed–Muller family that is foldable, encodable in $O(n \log n)$ over *any* sufficiently large field, and has proven minimum-distance bounds (e.g., relative distance ≈ 0.728 at message length 2^{25}, rate 1/8, 256-bit field). Third, and most important for compatibility, it **interleaves the folding IOPP with the sumcheck protocol**: because a fold with challenge α_i is a *partial evaluation* of the committed polynomial's multilinear extension, the d folding rounds can share their randomness with the d rounds of a sumcheck, so the committed object is a **multilinear** polynomial. This makes Basefold a drop-in PCS for sumcheck-based PIOPs (HyperPlonk, Spartan) that plain FRI cannot serve without an adapter, and it does so with $O(\log^2 N)$ proofs, more than 10× smaller than HyperPlonk-with-Brakedown at scale, and 2–3× faster than prior FRI-based multilinear PCS on the same field.

> **Evolutionary link.** Basefold answers *two* limitations at once. Against Brakedown/Ligero it answers the $\sqrt{N}$-proof limitation (folding gives polylog). Against FRI it answers the *univariate-only* limitation (sumcheck gives multilinear) and the *Reed–Solomon-only, field-bound* limitation (foldable codes give field agility). It is the structural convergence of the two lineages.

5.2 Basefold's soundness, and its tightening

Basefold's original soundness analysis is the sharpest illustration of the regime problem. Its *IOPP* soundness reaches a twice-applied-Johnson radius on the code's distance, inherited from the improved FRI analysis. But its *PCS knowledge-soundness and binding* (the property a commitment actually needs) are proven only in the **unique-decoding regime** ($\delta < \Delta_C/2$), because extraction there recovers a *unique* multilinear polynomial. The original paper also notes that round-by-round soundness (required to make the Fiat–Shamir compilation sound) is "straightforward" but does not prove it.

Haböck [10] later closes much of this gap for the Reed–Solomon instantiation, proving Basefold-over-Reed–Solomon sound up to the **Johnson list-decoding bound** by generalizing the correlated-agreement/proximity-gap theorem of [3] to linear *subcodes* of Reed–Solomon (plus a non-linear variant flexible enough to

also cover row-batched FRI-Binius and WHIR). Full list-decoding soundness for *general* random foldable codes (the field-agnostic case) remains open.

> **Systematization flag 3.** The Basefold most implementers cite as "sound in the list-decoding regime" is the Reed–Solomon instantiation analyzed by [10], not the general foldable-code scheme of [9], whose *committed-object* guarantee is proven at the unique-decoding radius. Conflating the two overstates what is proven for the field-agnostic construction.

5.3 WHIR: one protocol for both worlds, verified in microseconds

WHIR [8] is the more complete merger and the current frontier. Its object is a **constrained Reed–Solomon (CRS) code**: a Reed–Solomon subcode whose codewords additionally satisfy a sumcheck-type constraint $\sum_{b\in\{0,1\}^m} \hat{w}(\hat{f}(b), b) = \sigma$, where $\hat{w}$ is a *weight polynomial* and σ a target. This single algebraic object is expressive enough to encode *multilinear evaluation queries*, *univariate evaluations*, and *inner-product/linear claims*, so one proximity test answers the query types that previously required either FRI/STIR (univariate) or Basefold (multilinear) separately.

A WHIR round performs k rounds of sumcheck on the constraint (the Basefold idea) *and* a STIR-style rate-reducing fold on the code side (the STIR idea), unifying both mechanisms in one step. WHIR improves on STIR in two concrete ways: it adds the sumcheck/multilinear constraint capability, and it **removes an additive $O(\lambda^2)$ term** from the verifier's time. The latter is what produces WHIR's headline: verification in a *few hundred microseconds* (measured 210–770 μs across sizes 2^{18}–2^{30} and rates 1/2–1/16 **at the 100-bit security level**; at 128 bits the rate-1/2 verifier rises to ~ 0.9–$1.7\,ms$), where FRI, STIR, and Basefold verifiers run in milliseconds. Argument sizes span 36 KiB (rate 1/16, 100-bit) to 187 KiB (rate 1/2, 128-bit), verified against the paper's evaluation section. WHIR's soundness rests on **mutual correlated agreement**, a strengthening of correlated agreement in which, if a random linear combination agrees with the code on a large set S', then *every* individual word agrees on the *same* S'. Two radii must be distinguished to state WHIR's guarantee (this is the source of a common ambiguity). The *proximity* component (that the committed word is close to the constrained Reed–Solomon code) is provable up to the **Johnson bound** via the standard proximity-gaps toolbox [3,10], as for STIR. The new **mutual** correlated agreement property, which WHIR additionally needs to bind the sumcheck/multilinear constraint, was proven in the **unique-decoding regime** in the WHIR paper and *conjectured* up to capacity, the setting that drives its aggressive concrete parameters. Subsequent work of Haböck [11] extends mutual correlated agreement for Reed–Solomon codes to the Johnson bound, which brings WHIR's constraint knowledge-soundness for the Reed–Solomon instantiation up to Johnson as well; only the capacity setting remains conjectural, and its strongest form is now refuted (§5.4). This mirrors Basefold, whose list-decoding soundness for the Reed–Solomon instantiation was likewise supplied after the fact by [10] (§5.2).

Evolutionary link. WHIR answers STIR's two residual limitations: the $O(\lambda^2)$ verifier term (fixed → microsecond verification) and the univariate-only object (fixed → constrained Reed–Solomon subsumes multilinear). Its authors position it as "a direct replacement for FRI, STIR, Basefold, and others," and combined with a multilinear-PIOP compiler it yields an efficient SNARG for generalized R1CS.

5.4 The soundness axis, systematized

We can now state the paper's central thesis precisely. Across all seven schemes, the *asymptotic* query-complexity story is monotone and well understood: $\Theta(\lambda \log d)$ for FRI, dropping to $O(\log d + \lambda \log\log d)$ for STIR and WHIR, with the tensor schemes at $\sqrt{N}$ proofs orthogonally. But the *soundness regime* is neither monotone nor consistently reported, and it dominates concrete cost.

Table 2: Soundness regimes, and the practical query gap.

Scheme	Proven radius	Conjectured radius	Per-query soundness (proven)	Per-query (conjectured)
FRI	$\delta_0 \approx (1-3\rho)/4$; $1-\sqrt{\rho}$ via [3]	$1-\rho$ (refuted §5.4)	$\sim \frac{1}{2}\log\frac{1}{\rho}$ bits	$\log\frac{1}{\rho}$ bits
DEEP-FRI	$1-\sqrt{\rho}$	$1-\rho$ (refuted §5.4)	$\sim \frac{1}{2}\log\frac{1}{\rho}$	$\log\frac{1}{\rho}$
STIR	$1-\sqrt{\rho}$	$1-\rho$ (refuted §5.4)	$\sim \frac{1}{2}\log\frac{1}{\rho}$	$\log\frac{1}{\rho}$
WHIR	proximity: Johnson (std. toolbox); *mutual* corr. agr.: unique-dec.	$1-\rho$ (refuted §5.4)	$\sim \frac{1}{2}\log\frac{1}{\rho}$	$\log\frac{1}{\rho}$
Basefold	unique-dec. (orig.); $1-\sqrt{\rho}$ for RS [10]	capacity (refuted §5.4)	varies	n/a
Ligero / Brakedown	unique-dec. (code distance)	n/a	n/a	n/a

The concrete consequence is captured by ethSTARK's own parameter tables [13], reproduced in Table 3. At a fixed rate 1/4 and grinding budget, the *same protocol* needs 2–2.5× more FRI queries (and thus a 2–2.5× larger proof) when parameters are set to the *proven* Johnson analysis rather than the *conjectured* capacity analysis.

Table 3: ethSTARK FRI query count, proven vs. conjectured (rate 1/4, grinding $\zeta = 20$ bits).

Security λ	Conjectured queries s	Proven queries s	Ratio
80	31	79	2.5×
100	41	104	2.5×
128	55	140 (deg-4 ext.)	2.5×

Systematization flag 4 (live risk, now with a concrete outcome). Every "conjectured" entry above rests on a proximity-gap/correlated-agreement statement *up to capacity* $1-\rho$ that was, until recently, unproven, and in **November 2025 the strongest such conjectures were disproved.** Two independent works over the standard large-field regime are

central here. Crites and Stewart [24] disprove all three up-to-capacity conjectures: correlated agreement, **mutual correlated agreement** (the exact form WHIR relies on), and Reed–Solomon list-decodability up to capacity (the DEEP-FRI form). They also give the reduction from list-decoding to proximity gaps. Ben-Sasson, Carmon, Haböck, Kopparty, and Saraf [25] prove matching positive results *up to the Johnson radius* but show that "few-exception" proximity gaps *at or beyond* Johnson would require $\Omega(n^{1.99})$ exceptional combinations, i.e. the optimistic capacity form cannot hold with a small exception count. Three precise caveats keep this from being over-claimed: (i) **soundness up to the Johnson bound $1 - \sqrt{\rho}$ is unaffected**: everything proven in this paper stands; (ii) the refutation is established over large fields, and a *small-field* analogue, while a natural worry, is not something we can attribute to a specific result and we do not claim it; (iii) the *practical* impact reported by early analyses is modest, with estimates ranging from a few percent of usable proximity to at most a roughly 15% increase in proof size, because deployments can retreat to revised *up-to-list-decoding-capacity* targets (now the subject of an Ethereum Foundation "Proximity Prize"). The systematizing point survives intact and is in fact sharpened: the discount that essentially every deployed system takes by default (Section 6) was, in its strongest form, *wrong*, and the field is now repricing it. One caveat of our own standard applies here: [24], [25], and [11] are recent, not-yet-refereed preprints, so this reframing is contingent on them surviving review; the positive direction we rely on elsewhere, soundness up to the Johnson bound, does not depend on them.

5.5 The compilation layer: what IOP soundness does not give you

Every soundness statement in Sections 4–5 is about the *IOP*. Deployed systems ship *arguments*, obtained via the BCS/Fiat–Shamir compilation (§2.3), and four aspects of that compilation are central to concrete security yet routinely elided.

Round-by-round soundness. BCS soundness for a *multi-round* IOP does not follow from plain soundness; it requires **round-by-round (RBR) soundness** (equivalently, state-restoration soundness): at every partial transcript there is a well-defined "doomed" state from which the prover's success probability stays bounded regardless of future messages. FRI, DEEP-FRI, STIR, and WHIR are analyzed (in their strongest treatments) to be RBR-sound; **Basefold's original analysis proves plain soundness and only asserts RBR soundness as "straightforward"** (a gap later addressed by follow-up analyses). This matters because Fiat–Shamir is sound in the ROM *only* for RBR-sound IOPs, and the 2023–2025 wave of Fiat–Shamir attacks (on GKR/sumcheck-style protocols and on under-provisioned parameterizations) targets the systems where this property is assumed rather than proven. For a paper whose thesis is "proven versus trusted," RBR soundness is a second axis of the same phenomenon.

Grinding / proof-of-work. The concrete tables (Table 3) carry a "grinding" budget ζ (e.g., 20 bits) that is universal in deployed STARKs but rarely defined

in the theory papers. Grinding forces the prover to find a nonce whose hash has ζ leading zeros before the query phase, adding ζ bits of soundness at the cost of 2^ζ prover hashes, cheaply "buying" bits that would otherwise require more queries. It is a first-order concrete-soundness lever: ethSTARK's s counts in Table 3 are *net of* $\zeta = 20$ grinding bits, and removing grinding would raise every query count.

Batching. Real systems commit to many polynomials (trace columns, quotient, composition) and open them together. Correlated agreement (Theorem 2) is what licenses testing a *random linear combination* of all of them as a proxy for all individually; this is why the proximity-gaps toolbox, and not FRI itself, is the decisive result. Batching is where the n/q and n^2/q field-size terms of Theorem 1 enter concrete security, and it is where small fields (large n/q) are most dangerous.

Extraction in the list-decoding regime. The gap between IOPP soundness and PCS *knowledge*-soundness (visible in Basefold, whose IOPP reaches a Johnson-type radius but whose extractor is proven only at unique decoding) is the difficulty of extracting a witness when the committed word has a *list* of nearby codewords rather than one. Below unique decoding, extraction reads off the single close codeword; in the list regime, the extractor must pin down *which* list element the prover is bound to, which is where out-of-domain (DEEP) sampling and the capacity conjectures do their work. This is the technical locus where "proven versus conjectured" actually bites a commitment scheme, as opposed to a bare proximity test.

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6 Deployment: Theory Meets Engineering

The taxonomy and narrative pay off here. We map production and near-production systems to schemes (Table 4) and argue that two coupled variables (*field choice* and *soundness regime*) explain the landscape.

6.1 The field–scheme coupling

The hash-based family's defining freedom is the field. KZG's pairing pins a system to a ~ 256-bit pairing-friendly curve (BN254, BLS12-381); the hash-based family can pick any field. Production systems exploit this aggressively: Goldilocks ($2^{64} - 2^{32} + 1$) in Plonky2, Boojum, and Polygon Miden; BabyBear ($2^{31} - 2^{27} + 1$) in RISC Zero and SP1 Turbo; KoalaBear ($2^{31} - 2^{24} + 1$) in Plonky3 and SP1 Hypercube; Mersenne-31 ($2^{31} - 1$) in the Circle-STARK systems; and binary towers in Binius. Smaller fields mean faster field arithmetic and better hardware utilization, the direct cause of the family's prover-throughput dominance.

The subtlety is *two-adicity*, and here a common misconception must be dispelled. Radix-2 FFT (the standard low-degree extension) needs a smooth power-of-two multiplicative subgroup, i.e. a large power of two dividing $p - 1$. **BabyBear and KoalaBear are specifically engineered to have this**:

$p-1 = 2^{27} \cdot 15$ for BabyBear (two-adicity 27) and $2^{24} \cdot (2^7-1)$ for KoalaBear (two-adicity 24). This is *why* RISC Zero, SP1, and Plonky3 run FFT-based encoding over them with no special construction. The FFT-unfriendly field is **Mersenne-31** ($p = 2^{31} - 1$, for which $p - 1 = 2 \cdot 3^2 \cdot 7 \cdot 11 \cdot 31 \cdot 151 \cdot 331$ has two-adicity only 1): its arithmetic is the fastest of the 31-bit primes because $2^{31} - 1$ is a Mersenne prime, but it has no usable smooth two-adic subgroup. This is why the **Circle STARK** construction [15] exists: it recovers an FFT structure over Mersenne-31 via the *circle group* $\{(x, y) : x^2 + y^2 = 1\}$, which has order $p + 1 = 2^{31}$ (for $p \equiv 3 \bmod 4$) and therefore supplies power-of-two evaluation domains. Circle STARK now underpins StarkWare's Stwo (live on Starknet mainnet, November 2025) and Plonky3's circle module. The lesson for the taxonomy is that "small field" and "FFT-friendly" are *independent* knobs: Goldilocks/BabyBear/KoalaBear are both; Mersenne-31 is small-but-not-two-adic (rescued by the circle); binary-tower fields (Binius) are neither, and abandon FFT entirely in favor of additive-NTT / basis-change encoding.

6.2 The deployment map

Table 4: Production and near-production systems mapped to schemes.

System	Scheme	Field	Soundness setting	Rationale
ethSTARK (reference/doc)	FRI + DEEP-ALI	∼62-bit prime, ext.	**both regimes published**	the one system with a full proven-vs-conjectured spec (Table 3)
Stone (StarkWare, Cairo/Starknet)	FRI + DEEP-ALI	**252-bit Cairo prime** $2^{251}+17 \cdot 2^{192}+1$	conjectured †	native Cairo `felt252` field, a large-field FRI counterexample (see below)
Stwo / S-two (Starknet, Nov 2025)	Circle-FRI	Mersenne-31	conjectured ∼100b †	M31 fastest 31-bit field; circle group restores FFT
Plonky2 (Polygon)	PLONK + FRI	Goldilocks	conjectured ∼100b †	64-bit + FRI ⇒ cheap recursion, ∼ 40× vs. KZG
Plonky3 (Polygon)	AIR + FRI (+Circle)	BabyBear/KoalaBear/M31	conjectured ∼100b †	field-agnostic toolkit; pick fastest field
RISC Zero	FRI (DEEP-ALI) + Groth16 wrap	BabyBear (end-to-end)	conjectured ∼100b †	31-bit fits a 32-bit word; Groth16 only for on-chain
Boojum / zkSync Era	PLONK + FRI	Goldilocks	∼100b, boostable	Plonky2-like; 16 GB prover for decentralization
SP1 Turbo (Succinct)	Plonky3 AIR + FRI	BabyBear	conjectured ∼100b †	RISC-V zkVM; SNARK-wrapped
SP1 Hypercube (Succinct, mainnet 2026)	**sumcheck + LogUp-GKR, Jagged PCS** (multilinear)	**KoalaBear**	conjectured †	**the univariate→multilinear merger, deployed**: real-time ETH block proving
Polygon Miden	AIR + FRI	Goldilocks	conjectured ∼100b †	STARK zkVM; RPO hash
OpenVM (Axiom)	Plonky3 AIR + FRI	BabyBear	conjectured ∼100b †	modular zkVM framework

System	Scheme	Field	Soundness setting	Rationale
Ligetron (Ligero Inc.)	**Ligero** (tensor)	RS/linear	n/a	low-memory, browser/edge, no trusted setup
Expander (Polyhedra)	GKR + **Brakedown/Orion**-style code PCS	binary/small	n/a	linear-time GKR prover (integrated by Nexus zkVM)
Binius (Irreducible)	**FRI-Binius** (Basefold-style)	binary towers	n/a	no embedding overhead; tiny binary fields
Jolt (a16z)	sumcheck + Dory → **Binius** roadmap	256-bit → binary	n/a	sumcheck-only; Binius natural fit
Linea (ConsenSys, *hybrid*)	**Vortex** (Ligero/tensor) inner + PLONK-KZG wrap	RS tensor → BN254	n/a	hash-based inner prover, KZG only for on-chain, the §6.3 pattern, tensor-line
Scroll/PSE/Taiko (*pure-KZG contrast*)	Halo2 + **KZG**	BN254	KZG binding (trusted setup)	native cheap EVM on-chain verification

† Marked entries: the ∼100-bit conjectured setting is the family default; unlike ethSTARK, these systems do not publish an ethSTARK-grade proven-vs-conjectured spec, so the value is our inference from family practice, an inference now materially affected by the [24,25] refutations (§5.4).

6.3 What the map shows

The lineage is deployed, in order of maturity, and the merger has now reached mainnet. FRI is everywhere in production (StarkWare's Stone/Stwo, Polygon's Plonky2/3 and Miden, RISC Zero, zkSync Boojum, SP1 Turbo, OpenVM). Ligero ships in Ligetron and, as the Vortex inner prover, in Linea; Brakedown/Orion-family codes ship in Polyhedra's Expander (integrated by the independent Nexus zkVM). The most significant 2025–2026 development for this paper's thesis is that **the univariate→multilinear merger it identifies is no longer confined to research**: Succinct's **SP1 Hypercube** [26] (mainnet 2026) abandons AIR+FRI for a sumcheck + LogUp-GKR prover with a multilinear "Jagged" polynomial commitment over KoalaBear, a Basefold-adjacent, sumcheck-native design proving real-time Ethereum blocks. FRI-Binius (Irreducible, Polygon partnership) and Jolt's Binius roadmap point the same way. What *does* remain in research/formal-verification integration is specifically **STIR and WHIR** as named protocols (e.g., a Lean blueprint for STIR+WHIR), but their defining sumcheck/multilinear-constraint idea is what the deployed sumcheck-GKR provers embody (these provers do not perform STIR-style rate reduction, which is a Reed–Solomon-codeword technique). The theory-to-deployment lag is thus shorter than the "roughly two years" one would guess from the papers' dates: the *ideas* deploy well before the *named protocols* do.

The hybrid pattern is the norm. Almost no FRI-based system verifies its final proof on-chain directly; a large FRI/STARK proof is expensive to verify in an Ethereum smart contract. Instead, systems prove fast with a small-field FRI prover and then *wrap* the final proof in a Groth16 (RISC Zero) or Halo2-KZG

(much of the zkEVM stack) SNARG whose pairing check is cheap on-chain. This "fast transparent prover, succinct pairing-based wrapper" pattern is the clearest real-world expression of the field–scheme trade: small fields win the prover, pairings win the last-mile verifier, and production systems refuse to choose.

A large field is not always the wrong choice: the Stone counterexample. The field-coupling thesis ("small fields win the prover") has an instructive exception in StarkWare's production Stone prover, which runs FRI over the **252-bit** Cairo prime because that is Cairo's native `felt252` field, fixed long before the small-field trend. Stone shows the coupling is a tendency, not a law: a large field costs prover throughput but *strengthens* soundness (the n/q and n^2/q terms of Theorem 1 shrink), and when the arithmetization is already committed to a field, that constraint dominates. The 31-bit convergence (BabyBear/KoalaBear/Mersenne-31) is a property of *new, field-flexible* zkVMs, not an iron law of hash-based proving, and, as O2/O4 note, the smaller the field the more exposed a system is to the soundness terms the capacity refutations concern.

The soundness regime is a deployed decision, and the decision just got repriced. ethSTARK is the one system with a fully public account of the choice, and it deploys the *conjectured* regime, accepting Flag 4's risk in exchange for roughly halving proof size (Table 3). The rest of the field follows suit by default. Until November 2025 this was a bet on an unproven conjecture; it is now a bet on a conjecture whose *strongest form has been disproved over large fields* [24,25] (§5.4). The practical consequence is bounded (deployments retreat to revised up-to-list-decoding targets at a modest proof-size cost, not a doubling; see §5.4), but the qualitative point is now concrete rather than hypothetical: the entire deployed hash-based ecosystem took a discount that, in its optimistic form, was wrong, and is now repricing it. A systematization that only reported asymptotic query complexity would miss what most determines how these schemes are parameterized in practice, and would miss the recent shift in the status of the underlying conjectures.

6.4 The dimensions the asymptotics hide: hash, hardware, recursion

The cost model of Section 2.3 (queries times per-path hash cost) makes a hash function invocation, not a field multiplication, the atomic unit of a hash-based prover. Three practical dimensions follow, none visible in Table 1's asymptotics, all first-order in deployment.

Hash-function choice is the largest prover-cost lever. Because the commitment is a Merkle tree over the encoded matrix, the prover computes millions of hash compressions, and the *choice* of hash trades native speed against recursion cost. **Arithmetization-friendly** hashes are defined by low-degree algebraic round functions: Poseidon and Poseidon2 (Plonky2/Plonky3, RISC Zero recursion, SP1), Rescue-Prime-Optimized (Polygon Miden), Griffin, and Monolith. They are expensive to compute natively (a Poseidon2 permutation is far slower than a Blake3 block) but cheap to *verify inside a circuit*, which is what recursion requires. **Bit-oriented** hashes (Blake3, Keccak, SHA-256) invert the

trade: fast native, but their bit operations blow up an arithmetic circuit, making them costly to verify recursively. Deployed systems therefore *mix*: RISC Zero uses Poseidon2 in the recursion/Merkle pipeline and SHA-256 as the Fiat–Shamir oracle; Plonky3 ships Poseidon2, Rescue, Monolith, Blake3, and Keccak and lets the integrator choose per layer. This single decision often dominates prover wall-clock more than the choice of low-degree test, and it is the hidden reason the "fast prover + KZG/Groth16 wrapper" pattern (§6.3) exists at all: the wrapper's job is to convert a proof whose in-circuit re-verification cost is dominated by hashing into one a pairing check settles cheaply. A one-line soundness caveat belongs here: recent Gröbner-basis cryptanalysis of the HADES/Poseidon design [27,28] has trimmed security margins and revised recommended round counts *without* producing a break, a reminder that arithmetization-friendly hashes are younger and less battle-tested than SHA-2.

Hardware acceleration is why small fields win in practice. The prover-throughput advantage of small fields is realized on GPUs and custom hardware: 31- and 64-bit field elements fit machine words and vectorize, and the NTT/Merkle workload maps to massively parallel hardware (Ingonyama's ICICLE, Cysic, Irreducible's binary-field hardware, zkSync's Airbender GPU prover, Boojum's consumer-GPU target). SP1 Hypercube's real-time Ethereum block proving is a *16× RTX 5090* result [26], not a single-core asymptotic. A survey that compares $O(n \log n)$ provers without noting that the constant is a GPU kernel misrepresents the deployed cost landscape.

Recursion-friendliness is a first-class design axis. "Cheap recursion" recurs as a rationale (Goldilocks in Plonky2; Boojum's 16 GB target; Ligetron's low-memory pitch) but is really a composite property: a small field, an arithmetization-friendly hash, and a low-degree test whose verifier is cheap to express in-circuit. FRI's logarithmic verifier arithmetic and STIR/WHIR's reduced query counts (fewer Merkle paths to re-hash in the recursive verifier) feed directly into recursion cost. This is why WHIR's microsecond verifier matters beyond benchmarks: a cheaper verifier is a cheaper *recursion step*, which compounds across the proof tree that every zkVM builds.

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7 Discussion and Open Problems

O1: Field-agnostic list-decoding soundness. Basefold-over-Reed–Solomon is now proven to the Johnson bound [10], but the field-agnostic random-foldable-code construction (the reason to use Basefold at all) retains only a unique-decoding *committed-object* guarantee. Closing this is the most important open problem for the field-agnostic branch.

O2: Repricing after the capacity refutations. The up-to-capacity statements deployments optimistically assumed were disproved over large fields in late 2025 [24,25] (§5.4): specifically, correlated agreement to capacity (Proximity-Gaps' Conjecture 8.4), mutual correlated agreement to capacity (WHIR's), and

Reed–Solomon list-decodability to capacity (DEEP-FRI's Conjecture 2.3). FRI's own capacity conjecture (Conjecture 3) is not among the three directly refuted statements but loses its support, since it is downstream of these. The open questions are now *quantitative and forward-looking*: what is the exact revised soundness target up to *list-decoding* capacity (the Ethereum Foundation "Proximity Prize" question); how much do concrete proofs actually grow once systems reprice (early estimates put the increase well below the naive $2\times$); and whether an analogous refutation holds over the *small fields* (BabyBear, Mersenne-31, binary towers) now dominant in deployment, currently unresolved and not something the large-field results establish.

O3: Round-by-round soundness and Fiat–Shamir. Several schemes (Basefold in particular) assert but do not fully prove the round-by-round soundness needed for a sound Fiat–Shamir compilation. Given recent attacks on Fiat–Shamir in related settings, tightening these is a correctness priority, not a formality.

O4: Small fields and the circle. The move to 31-bit and binary fields (Mersenne-31, BabyBear, binary towers) both improves prover throughput *and* weakens the field-size term n/q in every proximity-gap soundness bound (Theorem 1). Since the large-field capacity refutations [24,25] do not directly transfer, the interaction between small-field performance and small-field *soundness* (whether the surviving Johnson-bound guarantees are tight enough over $q \approx 2^{31}$, and whether a small-field capacity refutation exists) is the live research front, sharpened rather than settled by the 2025 results.

O5: Consolidation. WHIR credibly claims to subsume FRI, STIR, and Basefold. Whether the ecosystem consolidates onto a single constrained-Reed–Solomon primitive, or continues to maintain scheme diversity for engineering reasons (memory, field, recursion friendliness), will determine whether the taxonomy of this paper is a snapshot of convergence or of lasting plurality.

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8 Conclusion

The hash-based low-degree test has, in seven years, gone from a single protocol (FRI) with a modest proven soundness ceiling to a family that spans two merged lineages, commits to both univariate and multilinear polynomials, verifies in microseconds, and secures a large share of deployed zero-knowledge systems. We organized this family along a taxonomy of committed object, code class, and testing mechanism; traced the evolutionary chain in which each scheme answers a concrete limitation of its predecessor: DEEP-FRI the soundness ceiling, Brakedown field-dependence, Basefold the field-agnostic/succinct/multilinear gap, STIR the constant-rate query barrier, WHIR the verification cost and the univariate–multilinear split; and grounded the theory in a survey of production systems where field choice and soundness regime jointly explain the engineering.

Our central finding is that the proven-versus-conjectured soundness axis, not asymptotic query complexity, is the load-bearing and least-consistently-reported

dimension of this design space. It determines concrete proof sizes, it is the axis on which the community's folklore has most drifted from the primary sources (from the DEEP-fold-versus-DEEP-ALI conflation to the FRI/Johnson attribution), and it is where the field's deployed parameters rested on capacity conjectures whose strongest forms were *disproved over large fields in November 2025* [24], with a matching exception-count barrier from [25]. That refutation did not invalidate the schemes (Johnson-bound soundness is intact and the practical repricing is modest), but it converted our thesis from a cautionary framing into a documented event: the discount the ecosystem silently took was, in its optimistic form, wrong. This systematization is ultimately about what these schemes actually prove versus what they are trusted to do, and, as of this writing, about what the field has just learned it may no longer trust.

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